

; 2. Sa se citeasca cate un caracter de la tastatura, fara ecou. Daca este cifra, sa se tipareasca pe
; ecran, daca este caracterul '\$', se termina programul, in orice alta situatie se adauga caracterul
; intr-un buffer care se va tipari in final pe ecran.

assume ds:data, cs:code

data segment

buffer db 256 dup(?) ; allocate 256 bytes for non-digit characters

msg_final db 13,10,'buffer: \$'

data ends

code segment

start:

mov ax, data

mov ds, ax

xor si, si ; clear si before starting to increment it <=> mov si, 0

read:

mov ah, 08h ; read character without echo

int 21h ; execute interrupt => then result is stored in al

cmp al, 0

je read

cmp al, '\$' ; this is the exit character

je final_prog

cmp al, '0' ; add to buffer if it is not a digit, otherwise to console

jb add_buffer

cmp al, '9'

ja add_buffer

mov ah, 02h ; print a single character

mov dl, al

int 21h ; display the digit on the screen using the dl register

jmp read

add_buffer:

cmp si, 255

jae read

mov buffer[si], al

inc si

jmp read

final_prog:

mov buffer[si], '\$'

mov ah, 09h ; print the string

mov dx, offset msg_final

int 21h

mov dx, offset buffer

int 21h

mov ax, 4C00h

int 21h

code ends

end start

Pseudocode sequence:

INITIALIZE Data Segment
SET BufferIndex = 0

REPEAT
 READ Character from Keyboard

 IF Character is NULL (0) THEN
 CONTINUE loop
 IF Character is '\$' THEN
 EXIT loop
 IF Character >= '0' AND Character <= '9' THEN
 PRINT Character to screen
 ELSE
 IF BufferIndex < 255 THEN
 STORE Character into Buffer[BufferIndex]
 INCREMENT BufferIndex
 END IF
 END IF
UNTIL '\$' is detected

APPEND '\$' to Buffer
PRINT "buffer: " label
PRINT Buffer content
TERMINATE Program

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DOSBox 0.74-3-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX

C:\>tasm lab4.asm
Turbo Assembler Version 3.1 Copyright (c) 1988, 1992 Borland International

Assembling file: lab4.asm
Error messages: None
Warning messages: None
Passes: 1
Remaining memory: 474k

C:\>tlink lab4.obj
Turbo Link Version 3.0 Copyright (c) 1987, 1990 Borland International
Warning: No stack

C:\>lab4
12828283
buffer: data type aici buff
C:\>lab4
2347823
buffer: literele se pun in buffer
C:\>lab4
8293
buffer: cifrele se afiseaza in consola
C:\>
```

```
DOSBox 0.74-3-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX

Z:\>SET BLASTER=A220 I7 D1 H5 T6

Z:\>mount c ~/Desktop/tasm
Drive C is mounted as local directory /Users/zagreansonia/Desktop/tasm/

Z:\>c:

C:\>tasm lab4.asm
Turbo Assembler Version 3.1 Copyright (c) 1988, 1992 Borland International

Assembling file: lab4.asm
Error messages: None
Warning messages: None
Passes: 1
Remaining memory: 474k

C:\>tlink lab4.obj
Turbo Link Version 3.0 Copyright (c) 1987, 1990 Borland International
Warning: No stack

C:\>lab4
12828283
buffer: data type aici buff
C:\>
```